FOUNDATIONS OF PHONONS & GENERAL THEORY

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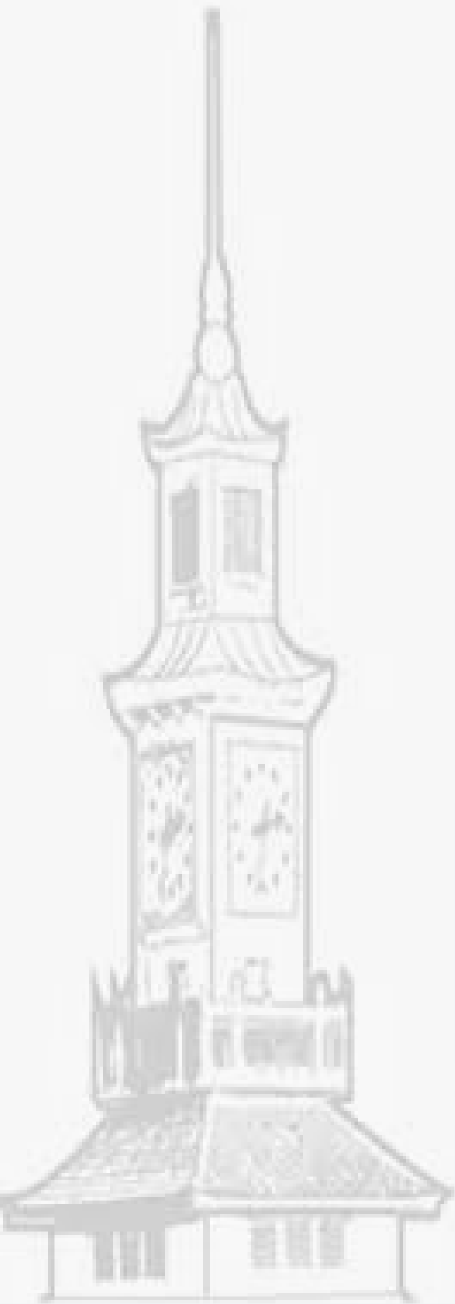
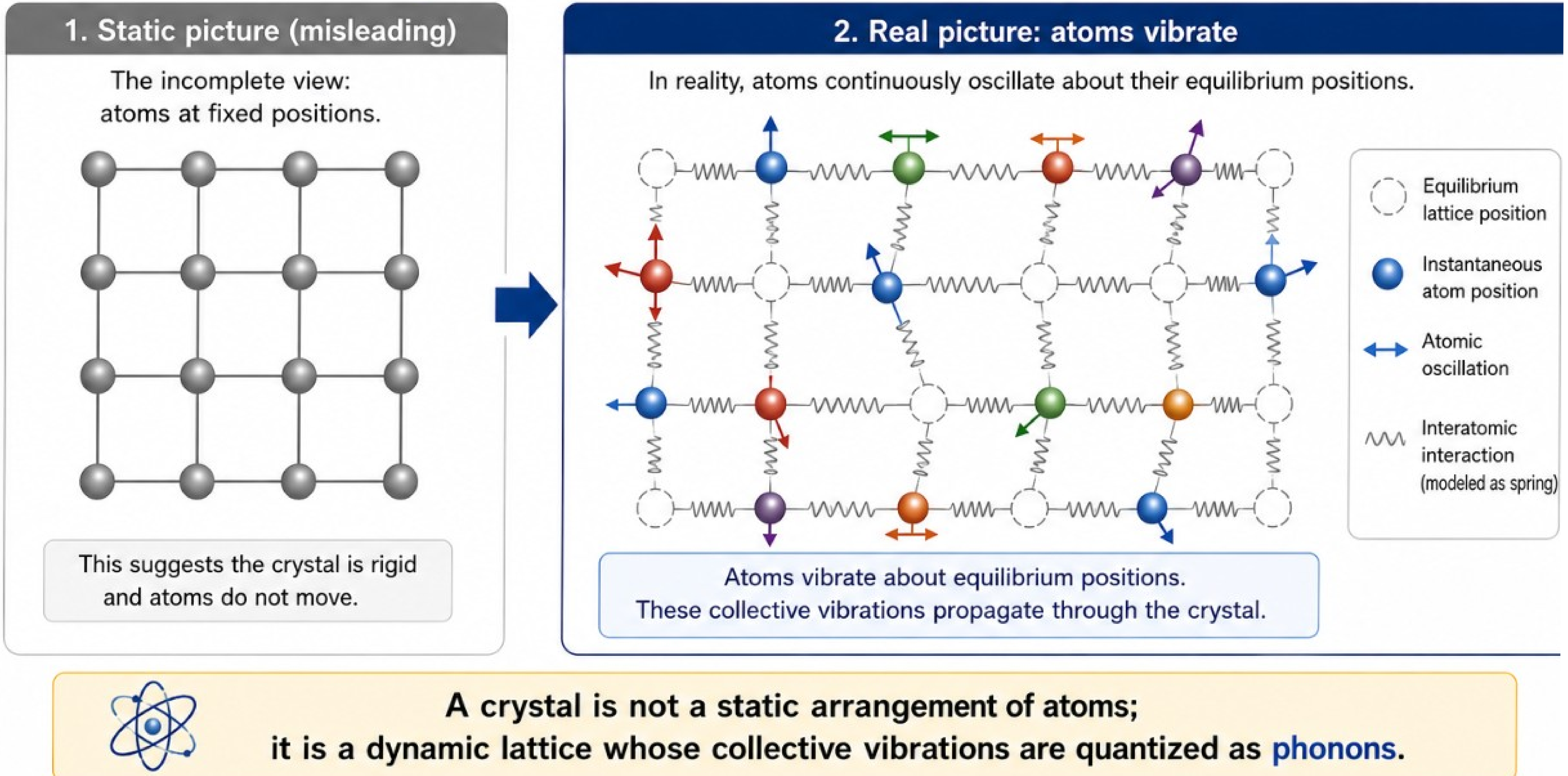


A Crystal: Static or Dynamic?



Lecture Learning Objectives

✓ Understand the foundational limits of the **harmonic approximation**.

✓ Master the explicit analytical derivation of **Phonon Dispersion Relations**.

✓ Formulate the physical and mathematical distinction between **Acoustic and Optical Phonons**.

✓ Understand how **phonons are computed from first principles** (Hands-On)

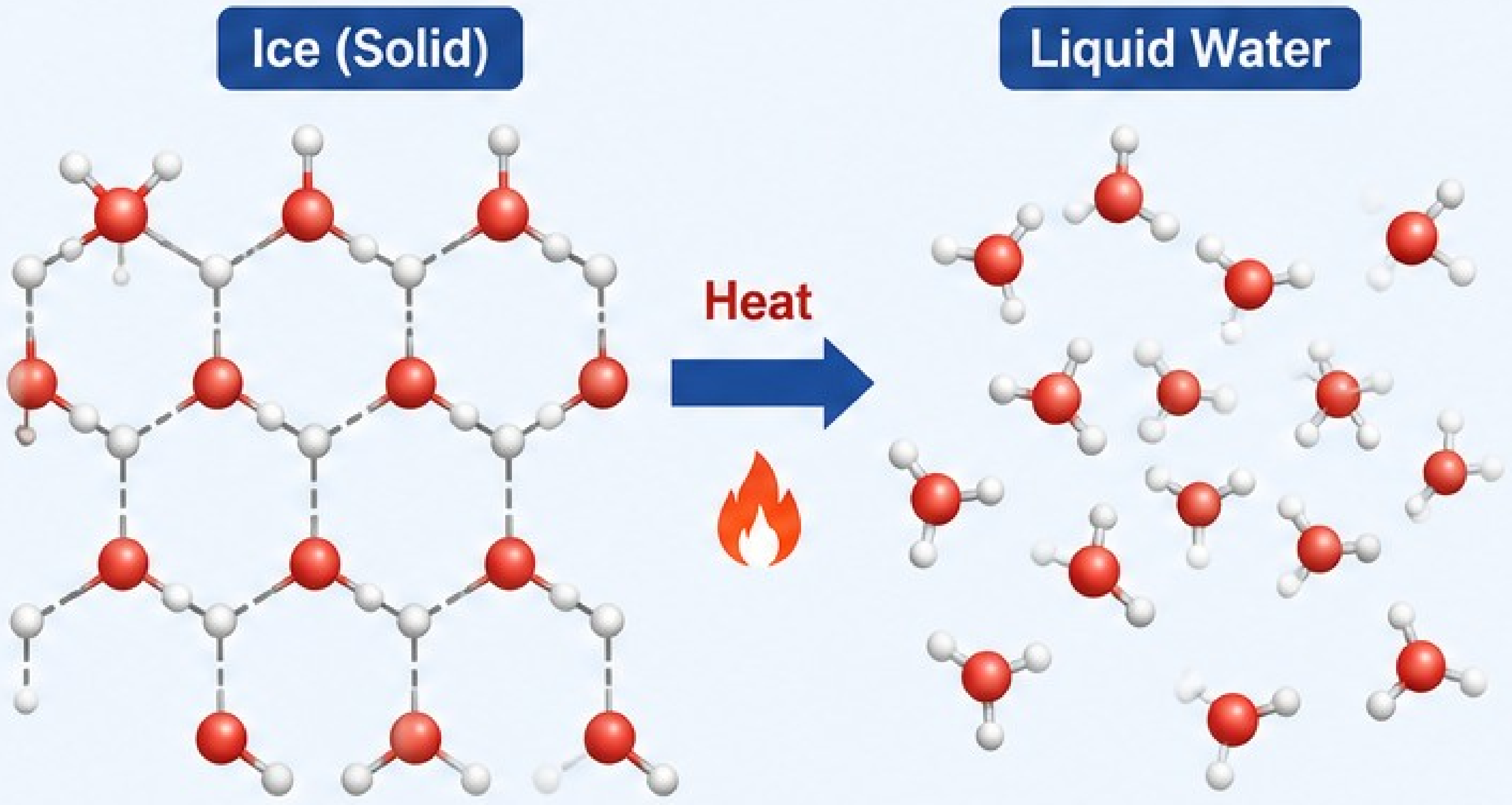
Why Should We Care About Phonons?

- **Thermal Fingerprints**: Determines structural **Heat Capacity** (C_v, C_p), **Vibrational Entropy** (S), and the **Helmholtz/Gibbs Free Energy** (F, G).
- **Transport & Scattering**: Controls Lattice **Thermal Conductivity** (κ) and mediates **electron-phonon** scattering governing **electrical resistivity** and conventional **superconductivity**.
- **Optical Spectroscopy**: Directly dictates Infrared (IR) Absorption lines and Raman Scattering peaks.
- **Structural Boundaries**: Dictates temperature-driven **structural phase transitions** and **lattice dynamic instabilities**.

The Microscopic Mechanism of Melting

The Question:

From a rigorous structural and energetic perspective, why does ice undergo a phase transition to liquid water when thermal energy is introduced?



Execution Strategy – Step 1:

Turn to your neighbor (2 minutes).

Execution Strategy – Step 2:

Formulate an explanation that links temperature directly to the mechanical stability of the lattice bonds.

The Microscopic Mechanism of Melting

The Question:

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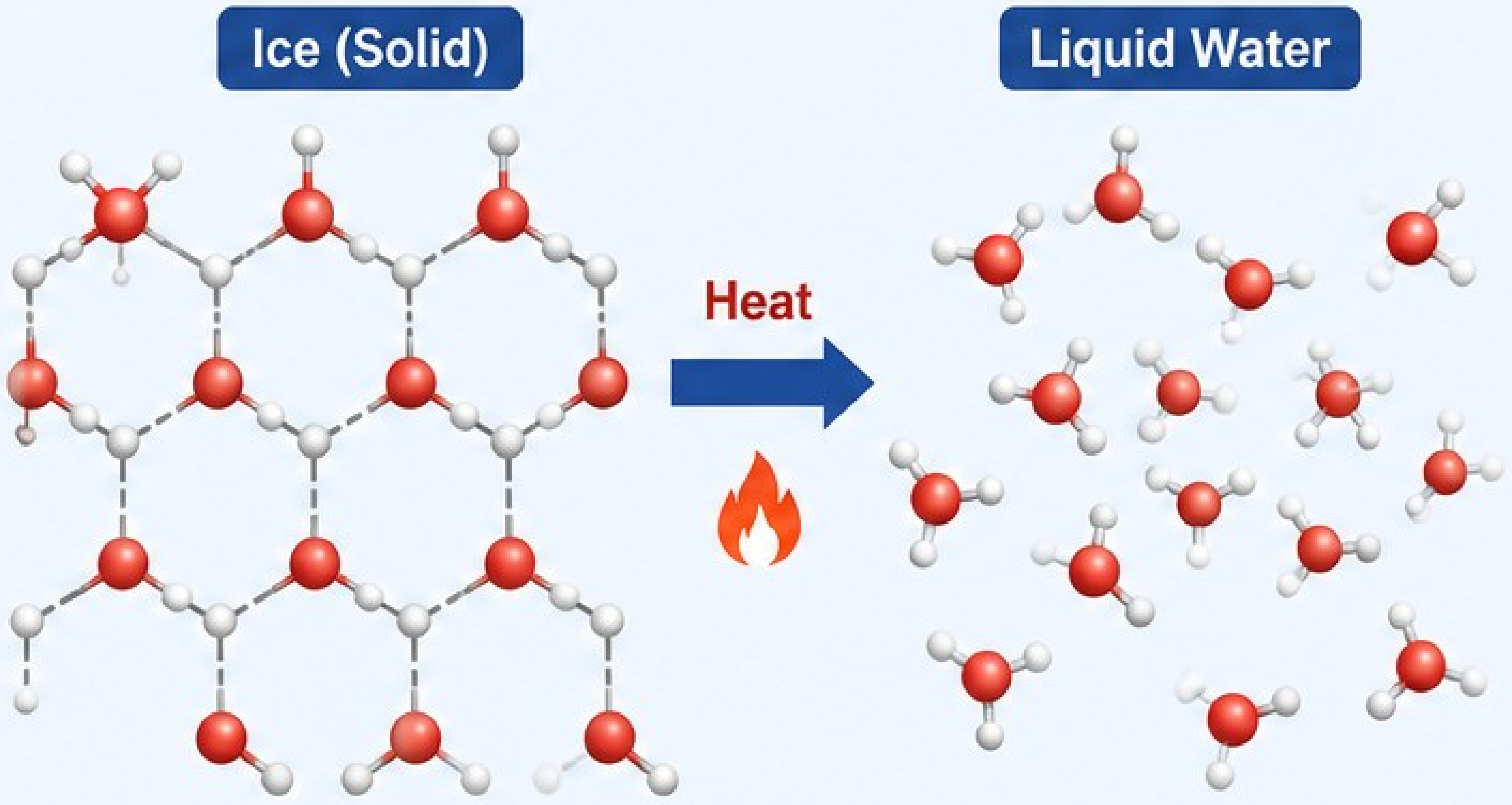
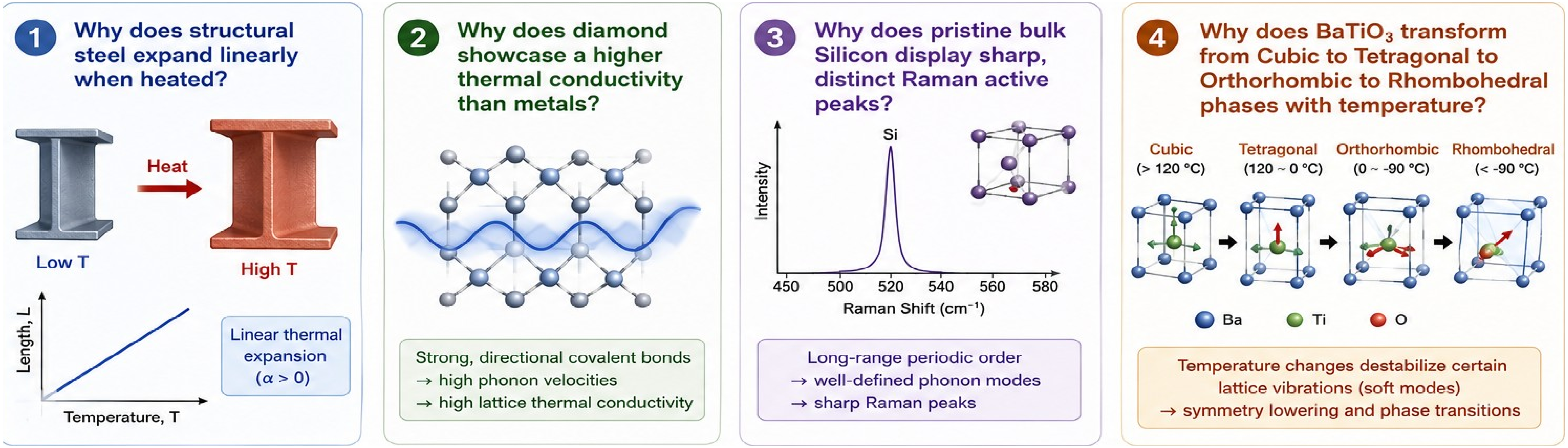


Think mechanistically, not just macroscopically.

Increasing temperature → Increasing population of vibrational modes → Rising mean-square displacement amplitudes → Breakdown of local restoring forces → Collapse of structural order (Lindemann melting criterion)

Macro-Observations Governed by Atomic Motion

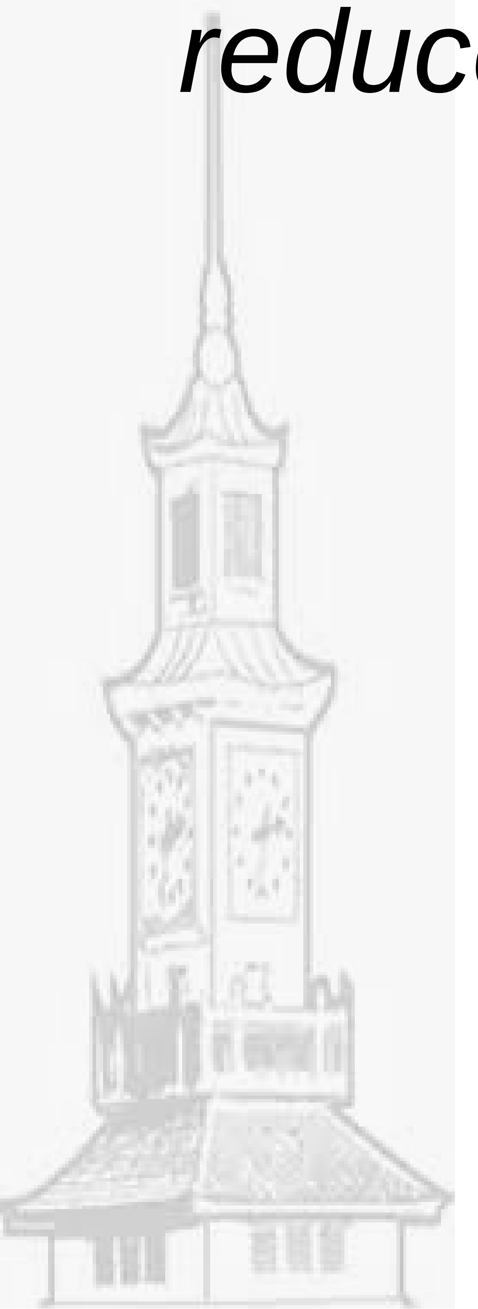


From thermal expansion to heat conduction, Raman spectra to ferroelectric phase transitions— all are governed by the dynamics of atoms: lattice vibrations and their interactions.

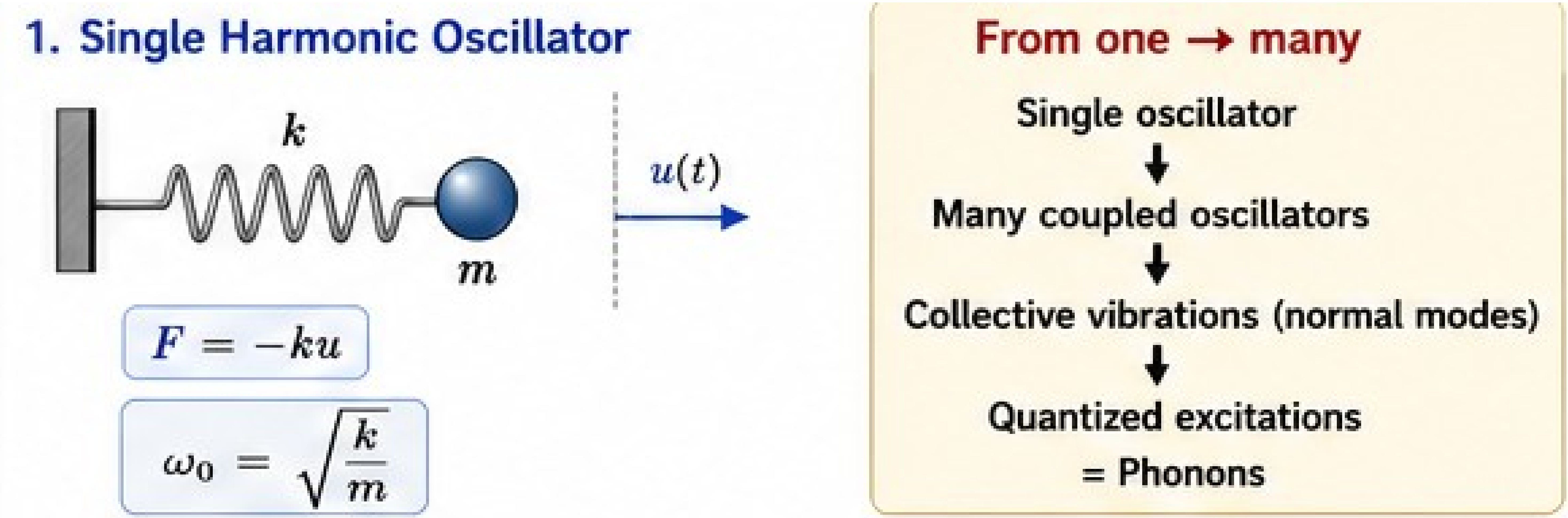
The Harmonic Paradigm

“Physics is a subset of human experience, which can be reduced to coupled harmonic oscillators.”

Michael Peskin

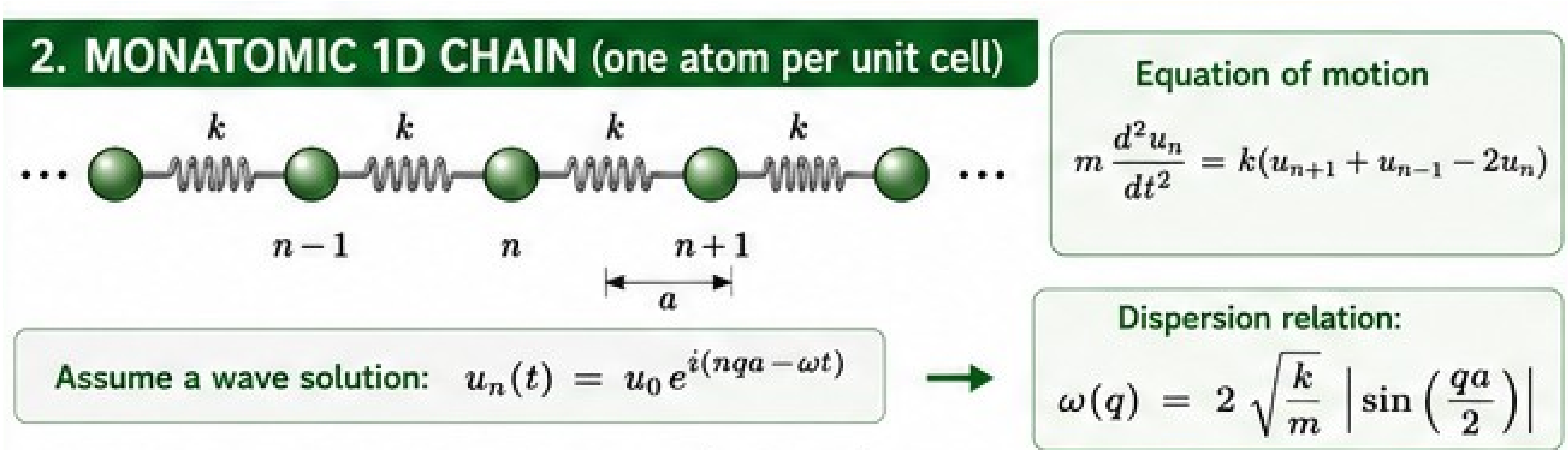


From a single Oscillator to Lattice Vibrations



Exr. Solve and obtain the frequency. What is the conditions that increase the frequency and relate it to the materials.

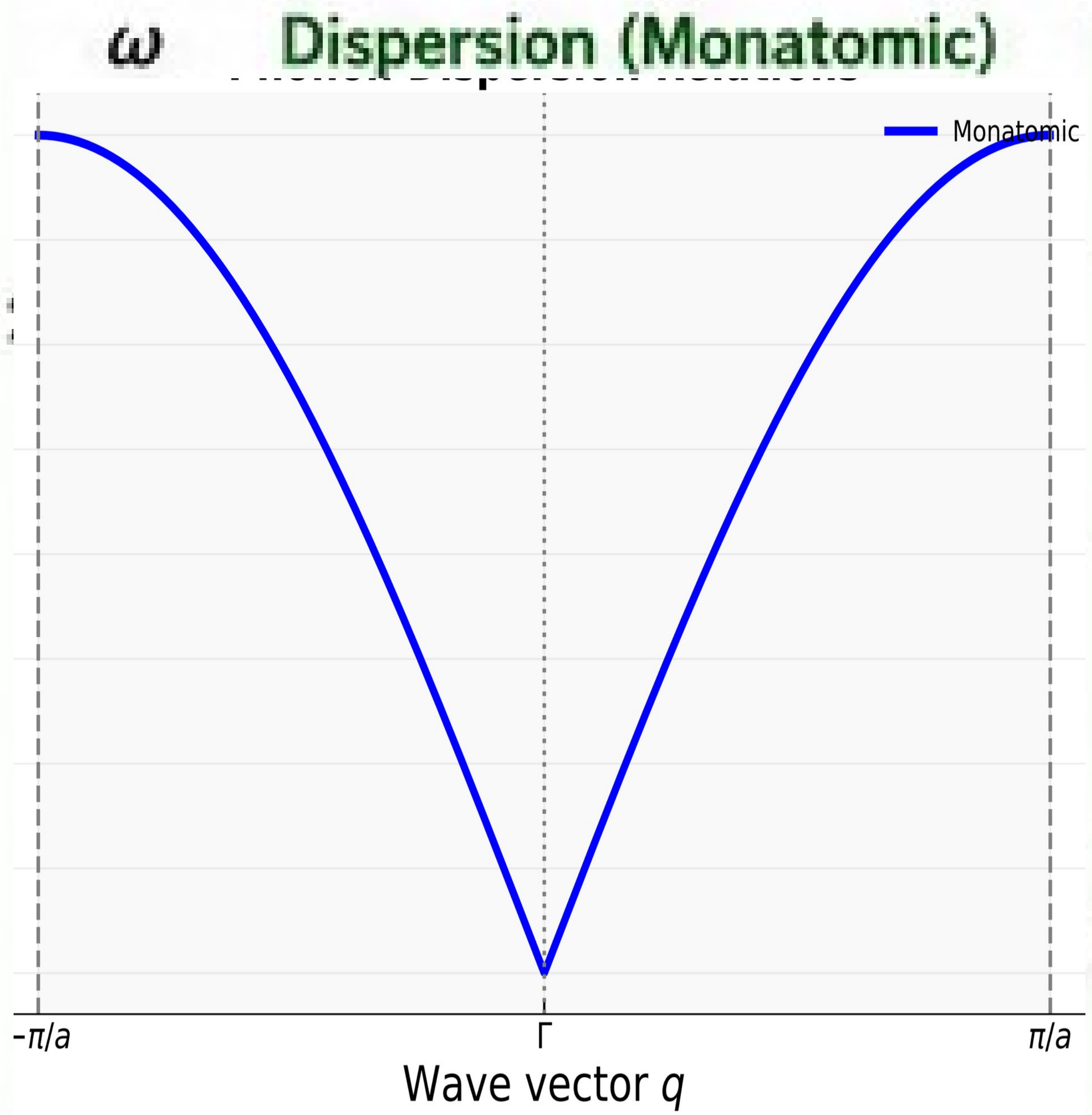
From a single Oscillator to Lattice Vibrations

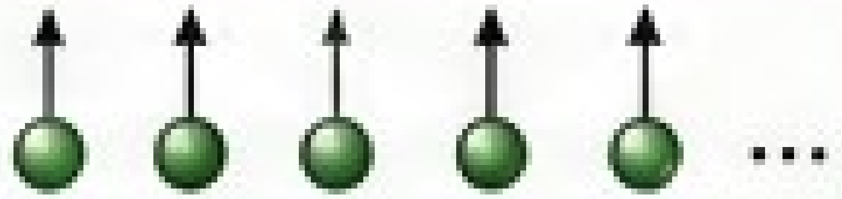
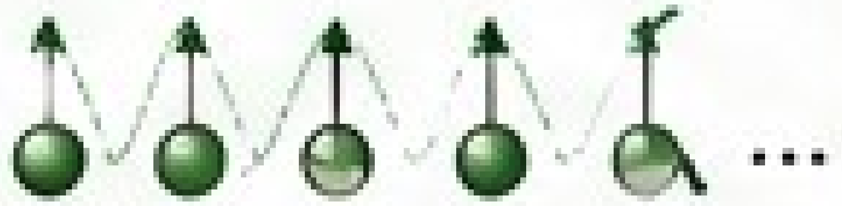
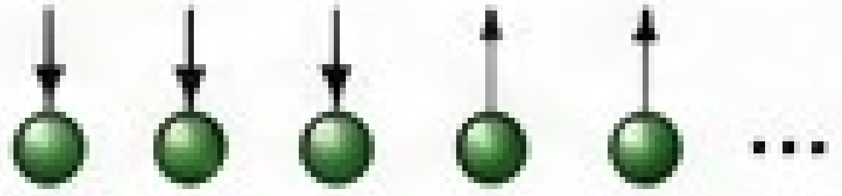


Exr. Solve and obtain the frequency (dispersion relation)

From a single Oscillator to Lattice Vibrations

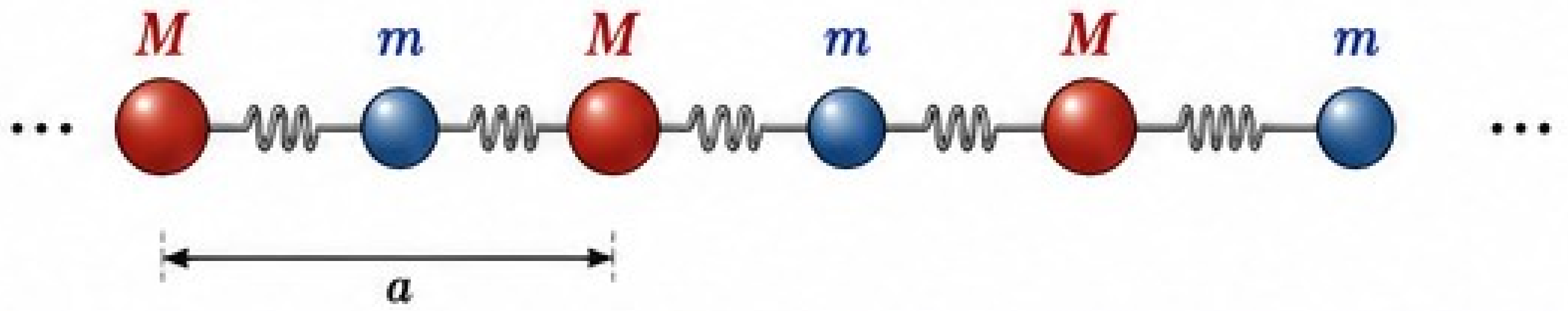
POSSIBLE EIGENVALUES (FREQUENCIES) AND CORRESPONDING DISPLACEMENTS



Mode (q)	Eigenvalue (Frequency)	Eigenvector (Displacement pattern) $u_n \propto e^{inqa}$	Physical Interpretation
$q = 0$	$\omega = 0$		All atoms move in phase (rigid translation)
$0 < q < \frac{\pi}{a}$	$0 < \omega < 2\sqrt{\frac{k}{m}}$		Wave-like collective motion
$q = \frac{\pi}{a}$	$\omega = 2\sqrt{\frac{k}{m}}$		Adjacent atoms move out of phase

From a single Oscillator to Lattice Vibrations

3. DIATOMIC 1D CHAIN (two atoms per unit cell)



Equations of motion

$$M\frac{d^2u_n}{dt^2} = k(v_n + v_{n-1} - 2u_n)$$

$$m\frac{d^2v_n}{dt^2} = k(u_n + u_{n+1} - 2v_n)$$

Assume wave solutions:

$u_n = Ue^{i(nqa-\omega t)}$ (atom M)

$v_n = Ve^{i(nqa-\omega t)}$ (atom m)

Dispersion relation (two branches):

$$\omega_\pm^2(q) = k\left(\frac{1}{M}+\frac{1}{m}\right) \pm k\sqrt{\left(\frac{1}{M}+\frac{1}{m}\right)^2 - \frac{4}{Mm}\sin^2\left(\frac{qa}{2}\right)}$$

Allowed wavevector: $q \in \left[-\frac{\pi}{a}, \frac{\pi}{a}\right]$

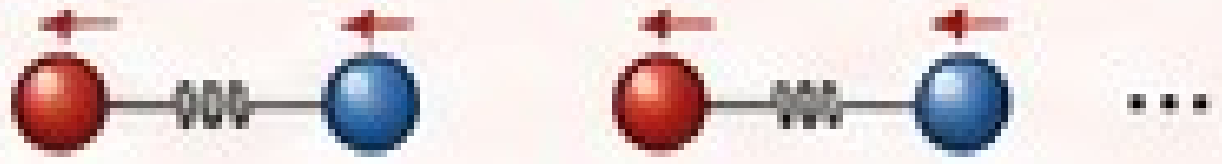
From a single Oscillator to Lattice Vibrations

POSSIBLE EIGENVALUES (FREQUENCIES) AND CORRESPONDING DISPLACEMENTS

Branch	q	Eigenvalue (Frequency)	Eigenvector (Displacement pattern)	Interpretation
Acoustic (ω_{-})	$q = 0$	$\omega = 0$		Atoms move in phase like a rigid translation
	$0 < q < \frac{\pi}{a}$	$0 < \omega < \sqrt{\frac{2k}{M}}$		Long-wavelength in-phase motion
	$q = \frac{\pi}{a}$	$\omega = \sqrt{\frac{2k}{M}}$		Neighboring atoms move out of phase

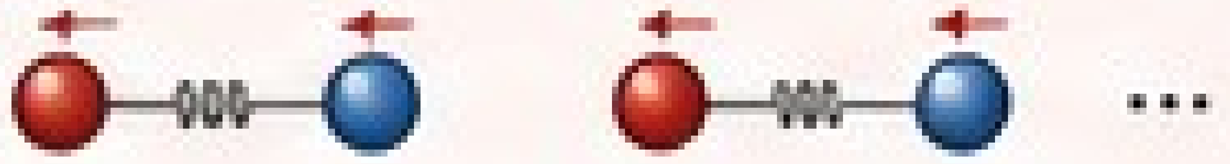
From a single Oscillator to Lattice Vibrations

POSSIBLE EIGENVALUES (FREQUENCIES) AND CORRESPONDING DISPLACEMENTS

Optical (ω_+)	$q = 0$	$\omega = \sqrt{2k\left(\frac{1}{m} + \frac{1}{M}\right)}$		Atoms move against each other (out of phase
	$0 < q < \frac{\pi}{a}$	$\sqrt{\frac{2k}{M}} < \omega < \sqrt{2k\left(\frac{1}{m} + \frac{1}{M}\right)}$		Optical-like out-of-phase motion
	$q = \frac{\pi}{a}$	$\omega = \sqrt{\frac{2k}{m}}$		Atoms strongly out of phase

From a single Oscillator to Lattice Vibrations

POSSIBLE EIGENVALUES (FREQUENCIES) AND CORRESPONDING DISPLACEMENTS

Optical (ω_+)	$q = 0$	$\omega = \sqrt{2k\left(\frac{1}{m} + \frac{1}{M}\right)}$		Atoms move against each other (out of phase
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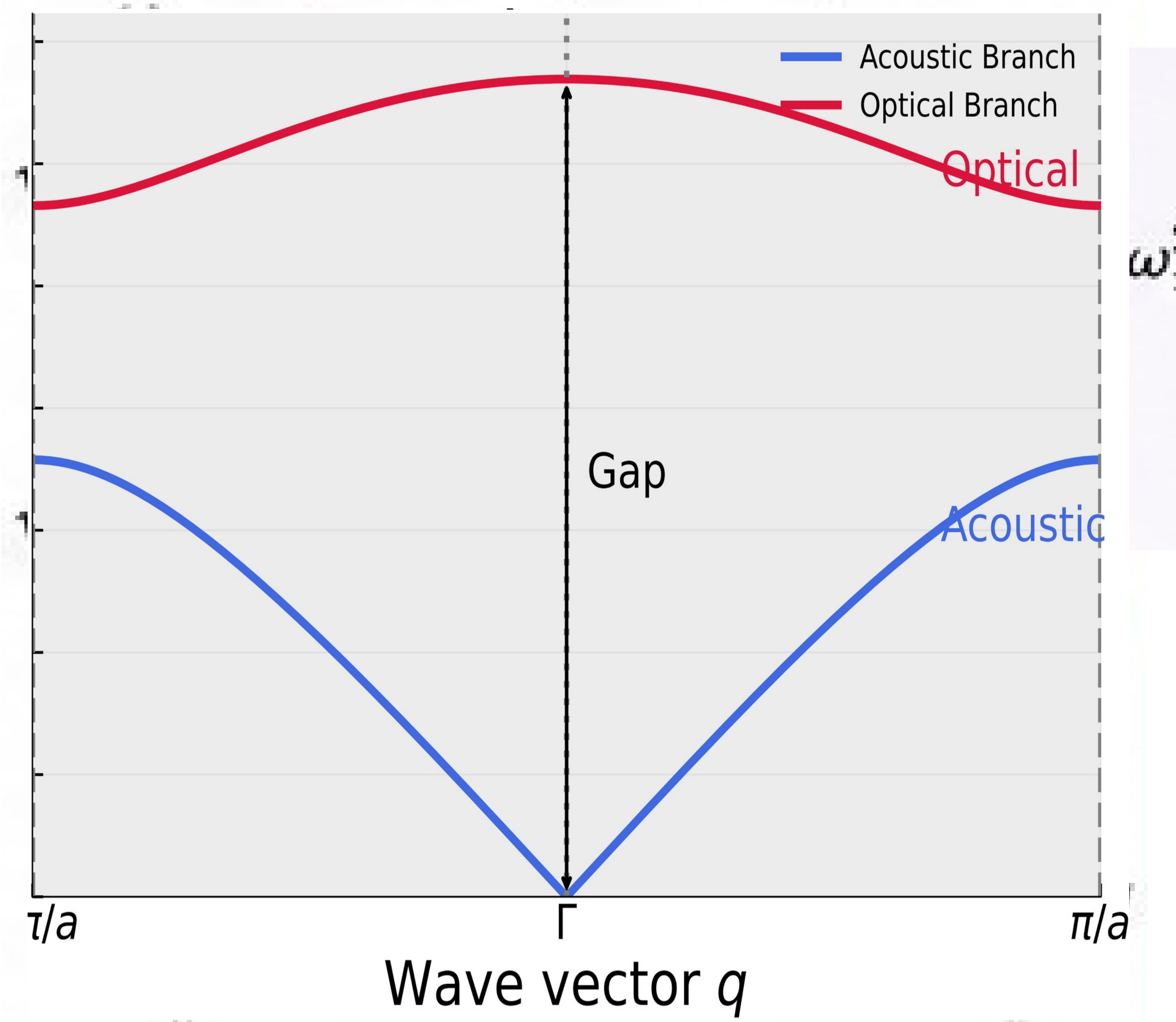
The optical branch has a finite frequency at $q = 0$, unlike the acoustic branch.

From a single Oscillator to Lattice Vibrations

POSSIBLE EIGENVALUES (FREQUENCIES) AND CORRESPONDING DISPLACEMENTS

Dispersion (Diatomic)



Dispersion relation (two branches):

$$\omega_{\pm}^2(q) = k\left(\frac{1}{M}+\frac{1}{m}\right) \pm k\sqrt{\left(\frac{1}{M}+\frac{1}{m}\right)^2 - \frac{4}{Mm}\sin^2\left(\frac{qa}{2}\right)}$$

Allowed wavevector: $q \in \left[-\frac{\pi}{a}, \frac{\pi}{a}\right]$

The optical branch has a finite frequency at $q = 0$, unlike the acoustic branch.

Back to Basics: the Born-Oppenheimer Approximation

$$\Psi_{tot}(\underline{r},\underline{R})=\psi_{el}(\underline{r};\underline{R})\chi_{NU}(\underline{R})$$

$$E_{tot}=E_{el}+E_{Nu}$$

Wavefunction of the system is factorized in to Electronic and Nuclear

$$\hat{H}_{el}\psi_{el}(\underline{r};\underline{R})=E_{el}\psi_{el}(\underline{r};\underline{R})$$

$$\hat{H}_{Nu}\chi_{Nu}(\underline{R})=E_{Nu}\chi_{Nu}(\underline{R})$$

$$\hat{H}_{el}=-\frac{1}{2}\sum_{i}^{N}\nabla_i^2+\sum_{i}^{N}\sum_{j>i}^{N}\frac{1}{r_{ij}}-\sum_{i}^{N}\sum_{\alpha}^{M}\frac{Z_\alpha}{r_{i\alpha}}$$

$$\hat{H}_{Nu}=-\frac{1}{2}\sum_{\alpha}^{M}\nabla_\alpha^2+\sum_{\alpha}^{M}\sum_{\beta>\alpha}^{M}\frac{Z_\alpha Z_\beta}{R_{\alpha\beta}}+E_{el}=-\frac{1}{2}\sum_{\alpha}^{M}\nabla_\alpha^2+V_{el}(\underline{R})$$

Towards vibrations: the Harmonic Approximation

For small deviations of nuclear coordinate, we can expand the potential energy in a Taylor series::

$$V(R) \approx V(R_0) + \left|\frac{\partial V}{\partial R}\right|_{R_0} \Delta R + \frac{1}{2!}\left|\frac{\partial^2 V}{\partial R^2}\right|_{R_0} \Delta R^2 + \frac{1}{3!}\left|\frac{\partial^3 V}{\partial R^3}\right|_{R_0} \Delta R^3 + \ldots$$

If we generalized for multidimensional, polyatomic case: :

$$V(\underline{R}) \approx V(\underline{R}_0) + \sum_{\alpha}^{M} \sum_{\mu}^{3} \left|\frac{\partial V}{\partial R_{\mu,\alpha}}\right|_{\underline{R}_0} \Delta R_{\mu,\alpha} + \frac{1}{2} \sum_{\alpha,\beta}^{M} \sum_{\mu,\nu}^{3} \left|\frac{\partial^2 V}{\partial R_{\mu,\alpha} \partial R_{\nu,\beta}}\right|_{\underline{R}_0} \Delta R_{\mu,\alpha} \Delta R_{\nu,\beta}$$

Let $\Delta R_{\mu,\alpha} = s_{\mu,\alpha}$ Then, the potential becomes

$$V(\underline{R}) \approx V(\underline{R}_0) + \sum_{\alpha}^{M} \sum_{\mu}^{3} \left|\frac{\partial V}{\partial R_{\mu,\alpha}}\right|_{\underline{R}_0} s_{\mu,\alpha} + \frac{1}{2} \sum_{\alpha,\beta}^{M} \sum_{\mu,\nu}^{3} \left|\frac{\partial^2 V}{\partial R_{\mu,\alpha} \partial R_{\nu,\beta}}\right|_{\underline{R}_0} s_{\mu,\alpha} s_{\nu,\beta}$$

Towards vibrations: the Harmonic Approximation

$$V(\underline{R}) \approx V(\underline{R}_0) + \cancel{\sum_{\alpha}^{M} \sum_{\mu}^{3} \left. \frac{\partial V}{\partial R_{\mu,\alpha}} \right|_{\underline{R}_0}} s_{\mu,\alpha} + \frac{1}{2} \sum_{\alpha,\beta}^{M} \sum_{\mu,\nu}^{3} \underbrace{\left. \frac{\partial^2 V}{\partial R_{\mu,\alpha} \partial R_{\nu,\beta}} \right|_{\underline{R}_0}}_{=\ \Phi,\ \text{Hessian Matrix}} s_{\mu,\alpha} s_{\nu,\beta}$$

= 0, because we are equilibrium position

= Φ, Hessian Matrix

$$\hat{H}_{Nu}(\underline{R}) = \frac{1}{2} \sum_{\alpha}^{M} \frac{1}{m_\alpha} \nabla_\alpha^2 + V(\underline{R}_0) + \frac{1}{2} \sum_{\alpha,\beta}^{M} \sum_{\mu,\nu}^{3} \Phi_{\alpha,\beta}^{\mu,\nu} s_{\mu,\alpha} s_{\nu,\beta}$$

Let's consider the classical EOM for each α nuclei:

$$F_{\alpha,\mu} = m_\alpha \frac{d^2 s_{\alpha,\mu}}{d t^2} = \sum_{\beta}^{M} \sum_{\nu}^{3} \Phi_{\alpha,\beta}^{\mu,\nu} s_{\nu,\beta} \longrightarrow$$

Time-dependent ansatz (nuclei are not frozen!)

Towards vibrations: The case of Periodic Systems

Time-dependent ansatz

$$F_{\alpha,\mu} = m_\alpha \frac{d^2 s_{\alpha,\mu}}{dt^2} = -\sum_{\beta}^{M} \sum_{\nu}^{3} \Phi_{\alpha,\beta}^{\mu,\nu} s_{\nu,\beta} \longrightarrow \boxed{s_{\mu,\alpha}(t) = u_{\alpha,\mu} e^{i\omega t}}$$

$$m_\alpha \omega^2 u_{\alpha,\mu} = \sum_{\beta}^{M} \sum_{\nu}^{3} \Phi_{\alpha,\beta}^{\mu,\nu} u_{\beta,\nu} \longrightarrow u_{\alpha,\mu} = \frac{1}{\sqrt{m_\alpha}} c_{\alpha,\mu} e^{i \underline{q} \underline{L}_n}$$

$\underline{L}_n$ is the Bravais lattice vector

$$\omega^2 c_{\alpha,\mu} = \sum_{\beta}^{M} \sum_{\nu}^{3} D_{\alpha,\beta}^{\mu,\nu}(\underline{q}) c_{\beta,\nu} \longrightarrow \omega^2 \underline{c} = \underline{\underline{D}}(\underline{q}) \underline{c}$$

Towards vibrations: The case of Periodic Systems

$$F_{\alpha,\mu}=m_\alpha \frac{d^2 s_{\alpha,\mu}}{dt^2}=-\sum_{\beta}^{M}\sum_{\nu}^{3}\Phi_{\alpha,\beta}^{\mu,\nu}s_{\nu,\beta} \longrightarrow$$

Time-dependent ansatz

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Towards vibrations: The case of Periodic Systems

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Two different ways to compute it:

- finite differences

*- linear response (DFPT)**

Towards vibrations: The case of Periodic systems

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*S. Baroni et al., Rev. Mod. Phys. 73, 515 (2001)

Towards vibrations: The case of Periodic Systems

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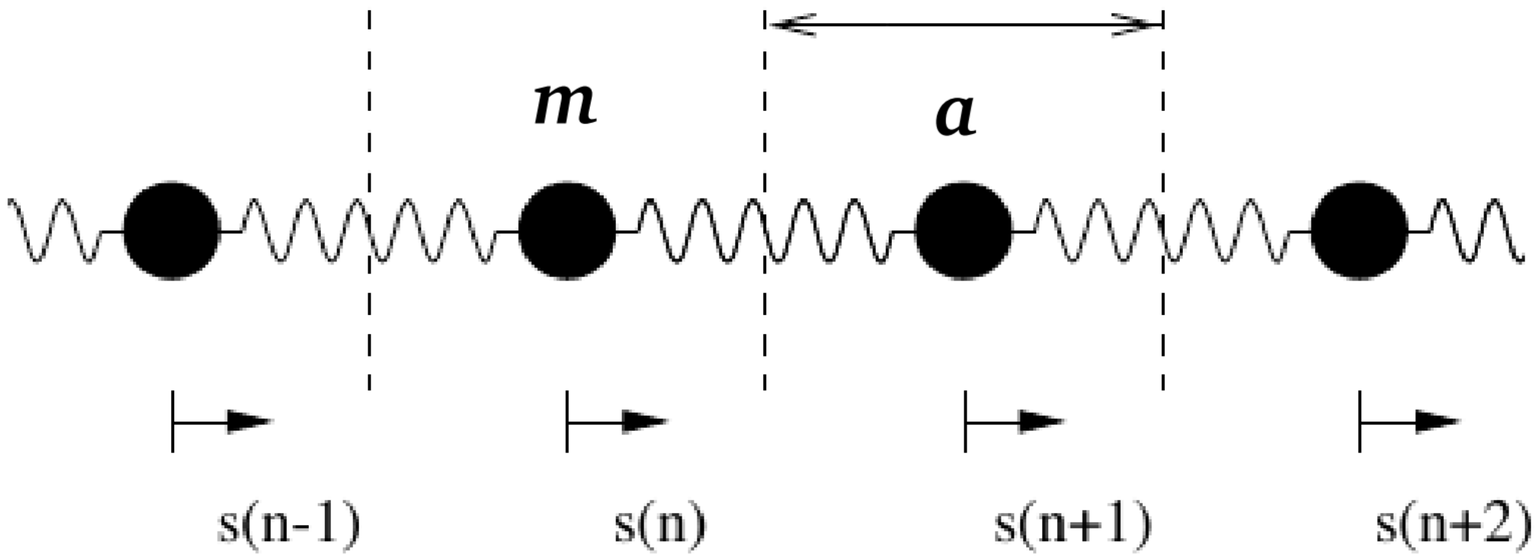
Solutions: phonon frequencies in $\underline{q}$-space!

$$\boxed{\omega = \omega_i(\underline{q}), i = 1, 2, \ldots, 3M}$$

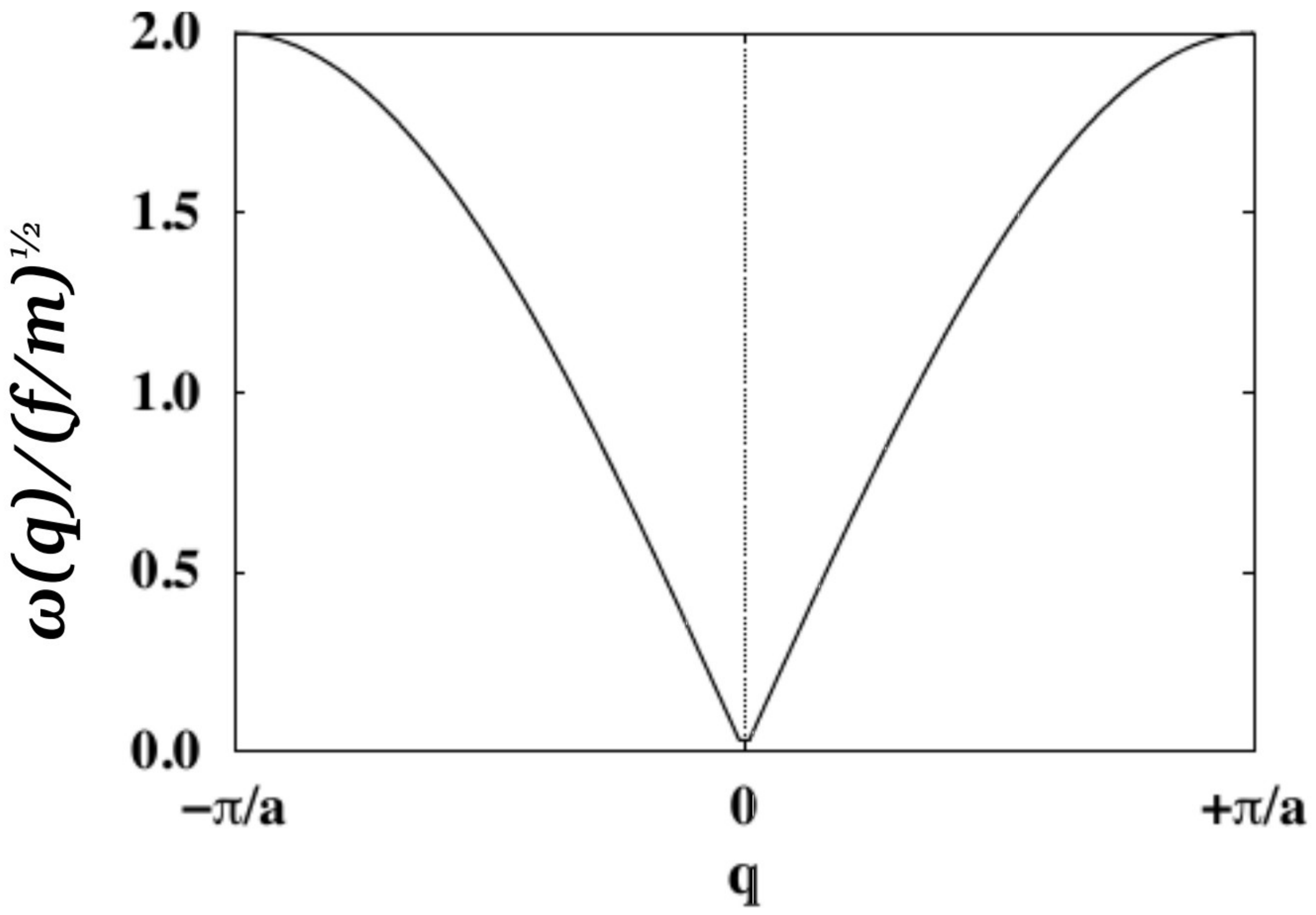
Towards vibrations: The case of Periodic Systems

- Linear chain with one atom per unit cell:



$$D(q)=\frac{f}{m}\left(2-e^{iqa}-e^{-iqa}\right)$$

$$\omega(q)=2\sqrt{\frac{f}{m}}\left|\sin\left(\frac{qa}{2}\right)\right|$$

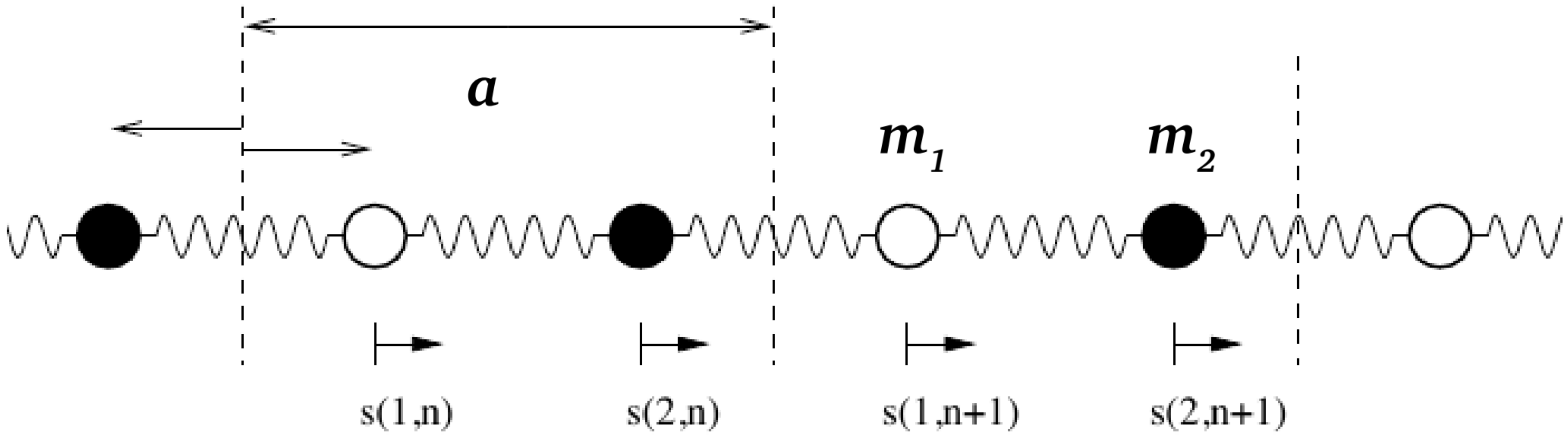


<u>Acoustic branches:</u>

- *For $q \to 0$, $\omega \approx aq(f/m)^{1/2}$: sound wave;*
- *Notable exceptions for some materials: graphene has a parabolic dispersion around Γ-point*

Towards vibrations: The case of Periodic Systems

- Linear chain with two different atoms per unit cell:



$$\omega(q)_{1,2}^{2} = f\left(\frac{1}{m_1}+\frac{1}{m_2}\right) \pm f\sqrt{\left(\frac{1}{m_1}+\frac{1}{m_2}\right)^2 - \frac{4}{m_1 m_2}\sin^2\left(\frac{q\,a}{2}\right)}$$

Towards vibrations: The case of Periodic Systems

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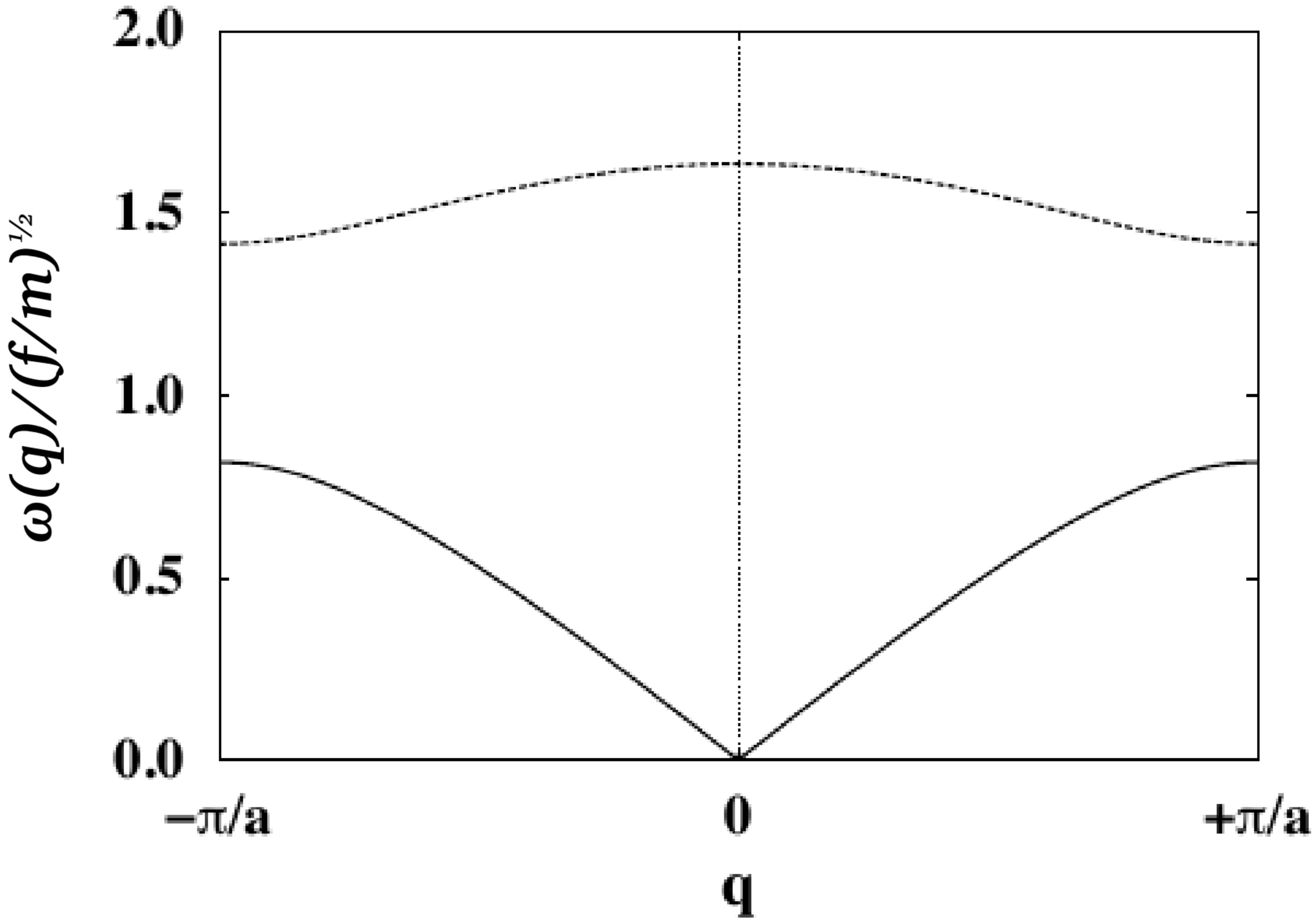
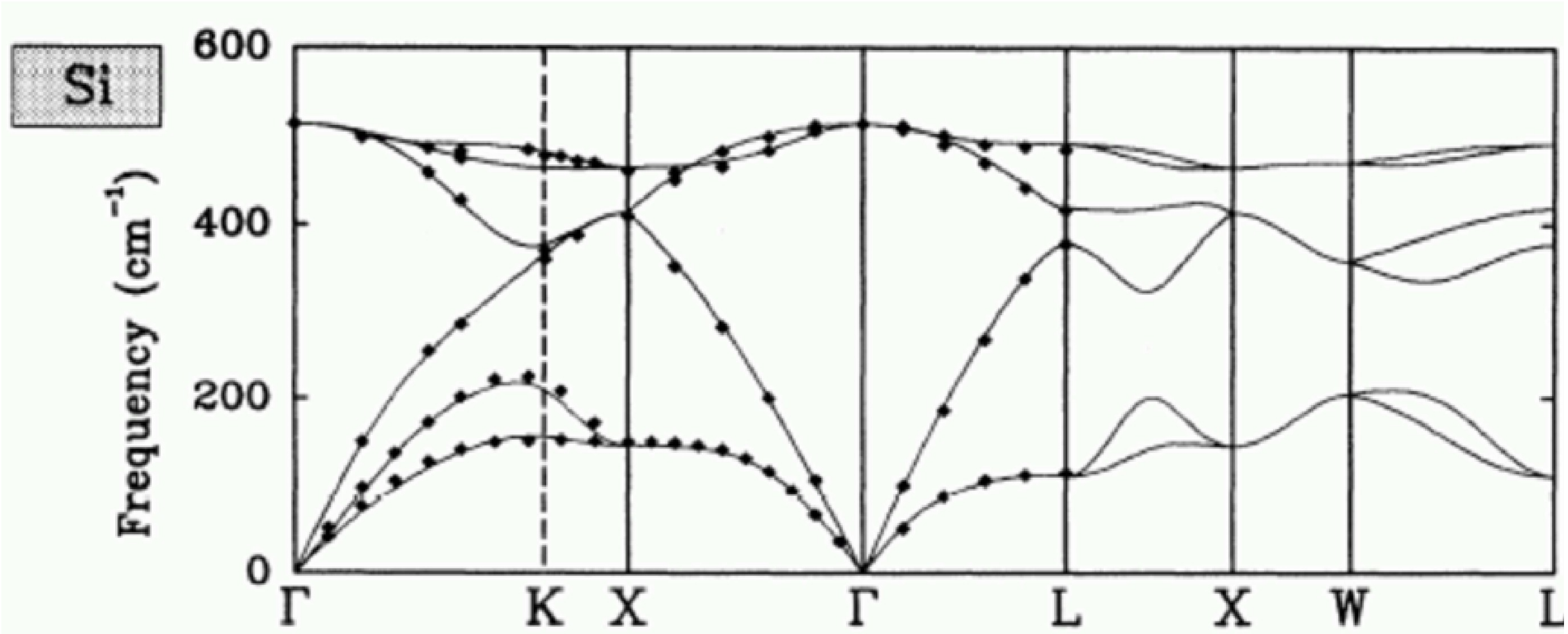


<u>Optical branches:</u>

- ***Both atoms in the unit cell vibrate against each other;***
- ***In ionic crystals these modes can be coupled to electromagnetic radiation***
- ***At the borders of the BZ we have a gap opening between acoustic and optical branches.***

Towards vibrations: The case of Periodic Systems

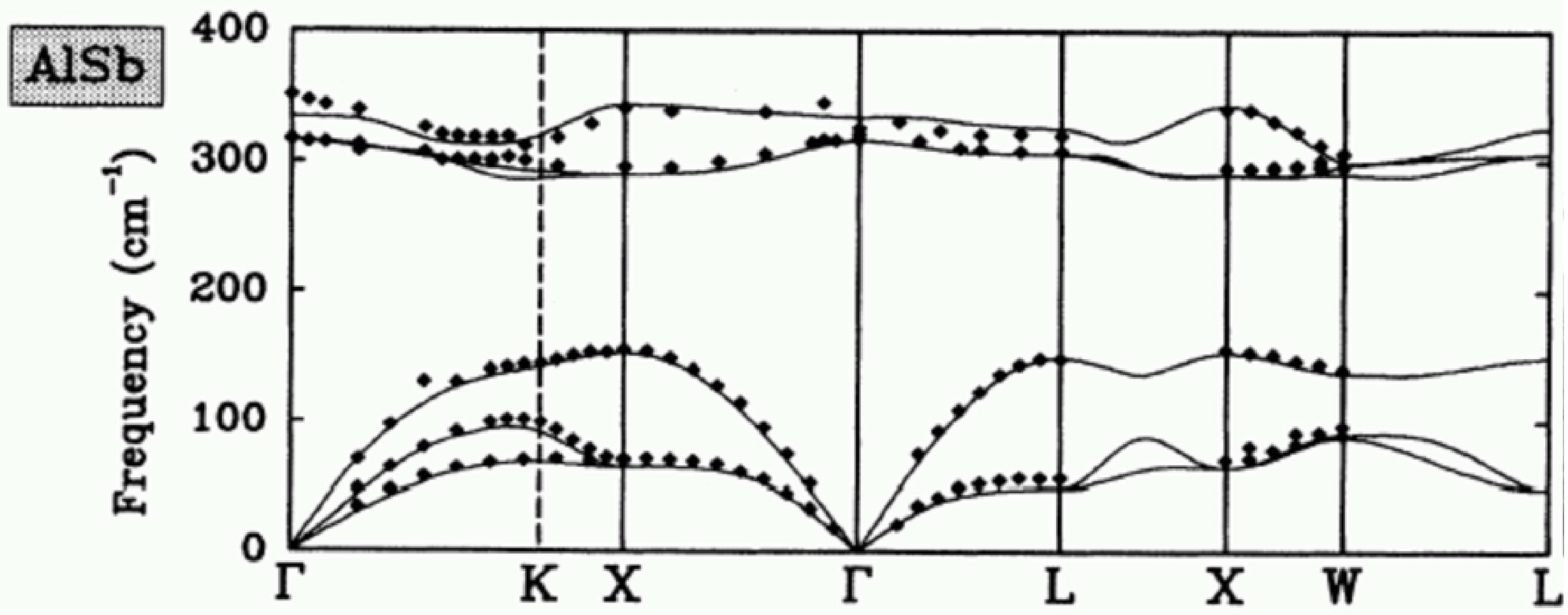
Calculation of phonon dispersion curves: A few examples



DFT-LDA vs experimental curves for bulk Silicon
(from P. Giannozzi et al. Phys. Rev. B, 43, 7231 (1991))

Towards vibrations: The case of Periodic Systems

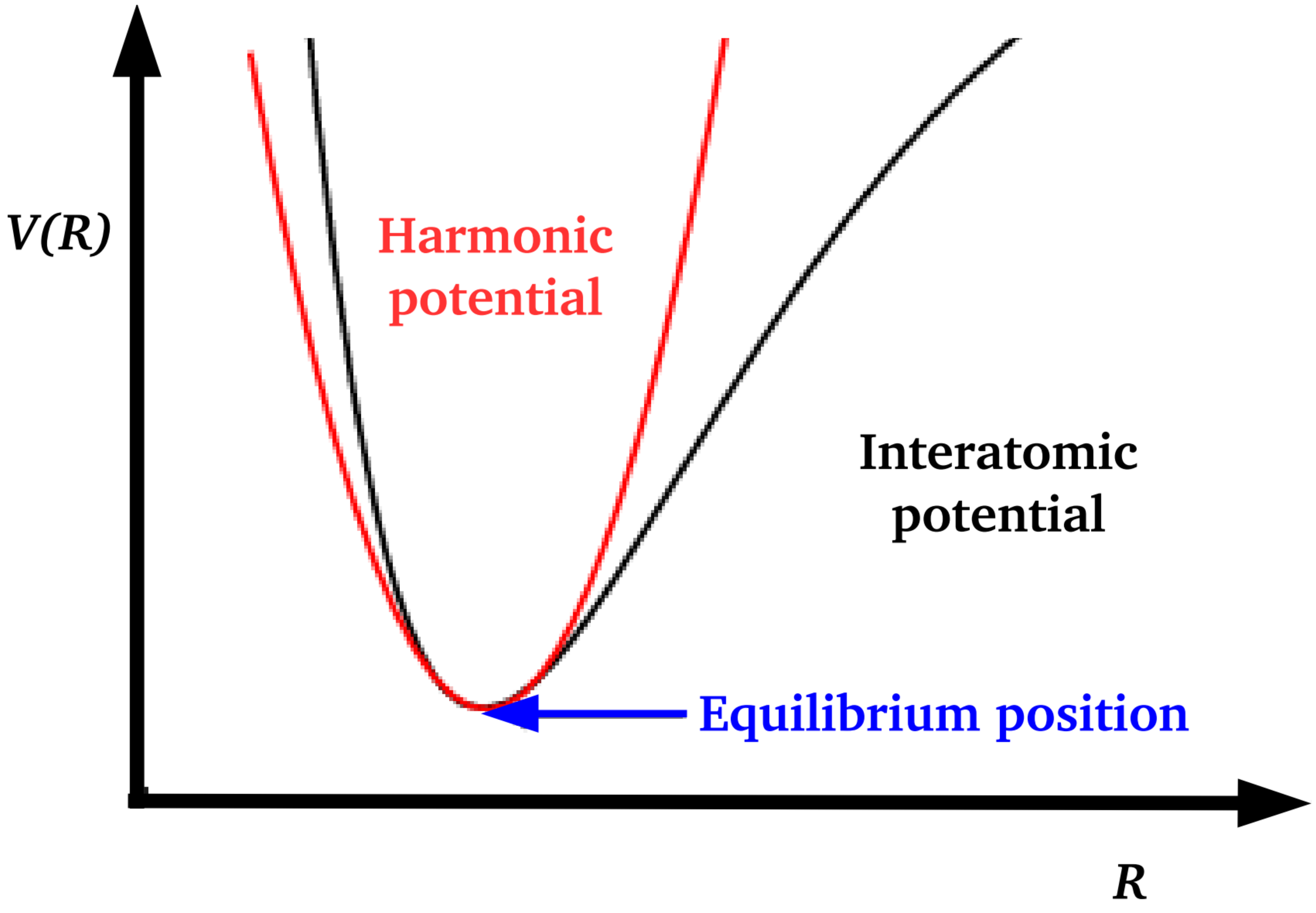
Calculation of phonon dispersion curves: A few examples



DFT-LDA vs experimental curves for bulk AlSb (same crystal structure as Si)
(also from P. Giannozzi et al. Phys. Rev. B, 43, 7231 (1991))

Towards vibrations: The case of Periodic Systems

Limitations of this approaches to get phonons:



- The harmonic approximation is only valid for small deviations from the equilibrium structure.
- Each mode is treated as independent from the others (i.e. no couplings).
- At high temperature, anharmonicity becomes important.
- For structures with many possible quasi-degenerate minima, imaginary modes can appear.
- Not suitable for non-equilibrium properties (like heat transport) or dynamical processes.

Thank you very much for your attention!

Next Hands-on